
What is Amazon SNS?

Amazon SNS is a **fully managed pub/sub (publish-subscribe) messaging service**. It allows **applications, services, or devices** to **send messages** to multiple recipients **at once**, without writing custom code for each recipient.

It is **event-driven** and helps in **decoupling microservices**, making systems more **scalable and fault-tolerant**.

Key Terminologies

Term	Meaning
Topic	A channel to which messages are published. Example: <code>OrderNotifications</code>
Publisher	A sender (e.g., an application, Lambda function) that publishes messages to a topic
Subscriber	A service or device (e.g., Email, Lambda, SQS, SMS) that receives messages
Protocol	The method used to deliver messages to the subscribers. Example: <code>email</code> , <code>sms</code> , <code>lambda</code>
Message	The content/data that is sent (e.g., alert, JSON data, text)

How SNS Works – Step-by-Step

Step 1: Create a Topic

A **Topic** is a central communication point to publish messages. You can name it something relevant like `MyOrderTopic`.

● Step 2: Add Subscriptions to the Topic

Once the topic is created, you add **subscriptions**, meaning who should receive messages.

✓ Options:

- Email
- SMS
- AWS Lambda (for auto-processing)
- Amazon SQS (for buffering)
- HTTPS endpoint (webhooks)

✉ For **email**, you must **confirm the subscription** via a confirmation email.

● Step 3: Publish Messages to the Topic

Now, any **application, AWS service (like Lambda), or user** can **publish messages** to the topic.

✓ When a message is published:

- It is **sent to all confirmed subscribers**
 - SNS handles the **fan-out pattern** — sending the same message to multiple endpoints
-

Real-World Student Example – SNS Lab

Scenario:

Students build an e-commerce app, and upon **successful order placement**, an SNS topic sends:

- An **email confirmation** to the buyer
 - An **SMS** to the delivery manager
 - A **Lambda** to trigger shipment logic
-

Demo Walkthrough: SNS Notification with Email

◆ Step 1: Create Topic

- Go to **AWS SNS Console**
- Click “**Create Topic**”
- Choose:
 - Topic Type: **Standard**
 - Name: **MyAlertTopic**
- Click “Create topic”

◆ Step 2: Create Subscription

- Inside the topic, click “**Create subscription**”

- Protocol: **Email**
- Endpoint: **your-email@example.com**
- Create it

➡ You will receive a **confirmation email**. Click the link to confirm.

♦ Step 3: Publish Message

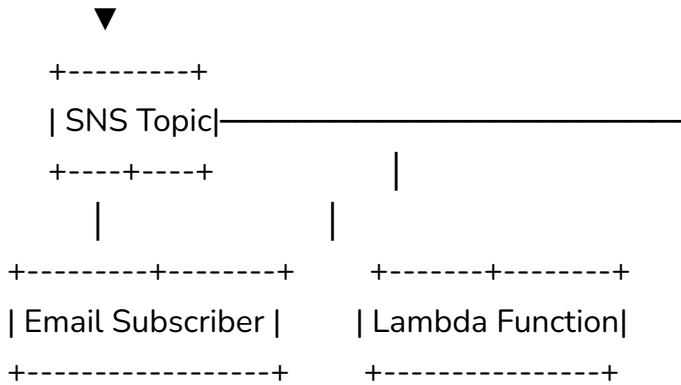
- Back in SNS console, go to the topic
- Click “**Publish message**”
- Enter:
 - **Subject:** **Test Notification**
 - **Message Body:** “This is a test alert from Amazon SNS!”
- Click **Publish**

📧 You will receive the email shortly.

🔄 SNS Fan-Out Pattern

Amazon SNS can **send one message to many services at once**:

```
+-----+
| Application |
+-----+-----+
|
Publishes Message
```



SNS Integrations

Integration	Use Case
Lambda	Auto-process messages
SQS	Queue messages for delayed or batch processing
Email	Notify users or admins
SMS	Urgent alerts
HTTP/S	Webhooks to 3rd party apps (like Slack, Zapier)

Security in SNS

- Use **IAM Policies** to allow/deny `sns:Publish`, `sns:Subscribe`, etc.
- Encrypt messages at rest using **KMS (AWS Key Management Service)**
- Enable **logging with CloudWatch** to track delivery status

Monitoring & Troubleshooting

- Use **CloudWatch Metrics** to see number of messages published and delivered
 - **Dead-Letter Queue (DLQ)** support for failed deliveries
 - **Retry policies** (e.g., exponential backoff for Lambda)
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Summary Table

Task	Action
Create Topic	Console / CLI / SDK
Add Subscription	Email, SMS, Lambda, SQS
Publish Message	Console, CLI, SDK
Confirm Subscriptions	Via email or endpoint
Monitor Deliveries	CloudWatch

CLI Example to Create and Publish SNS

Create a topic

```
aws sns create-topic --name MyAlertTopic
```

Subscribe an email

```
aws sns subscribe \  
--topic-arn arn:aws:sns:region:account-id:MyAlertTopic \  

```

```
--protocol email \  
--notification-endpoint student@example.com
```

Publish a message

```
aws sns publish \  
--topic-arn arn:aws:sns:region:account-id:MyAlertTopic \  
--subject "Lab Message" \  
--message "You've received this SNS test message!"
```
